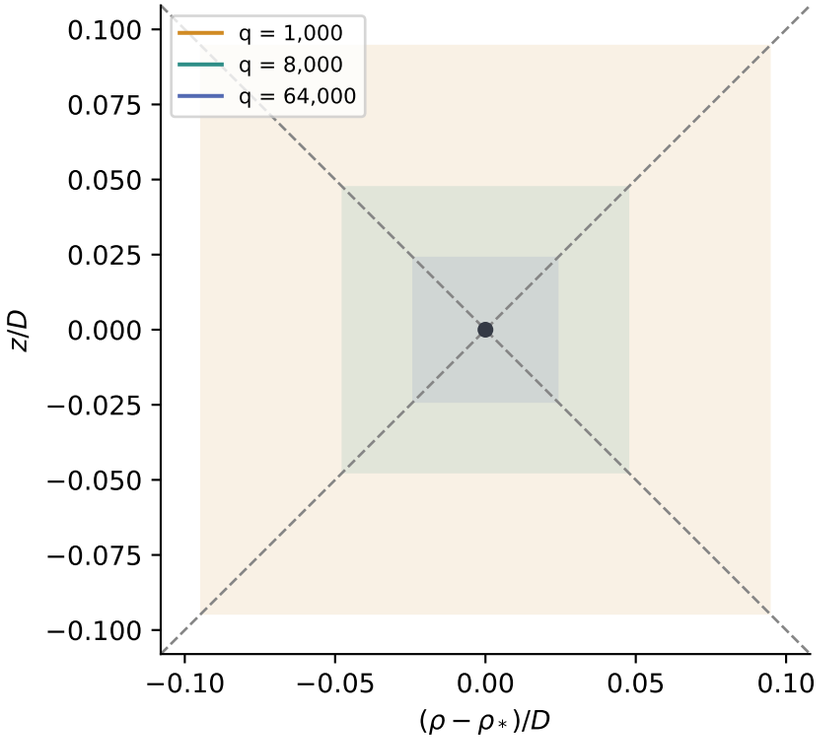
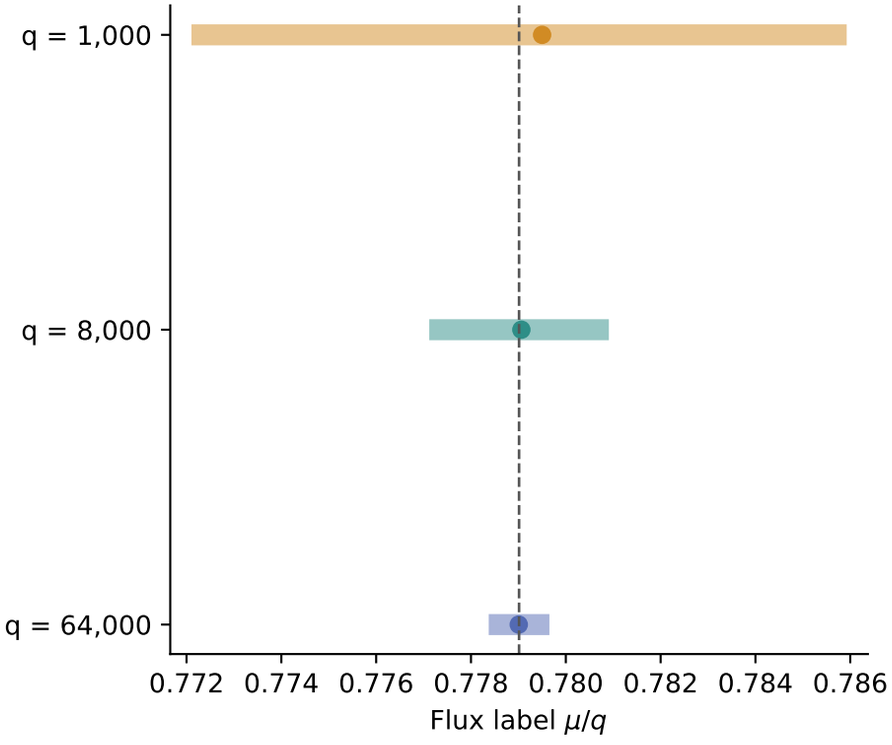
Where magnetic guiding breaks down | Local Dirac scaling at the null ring

Local mixing region



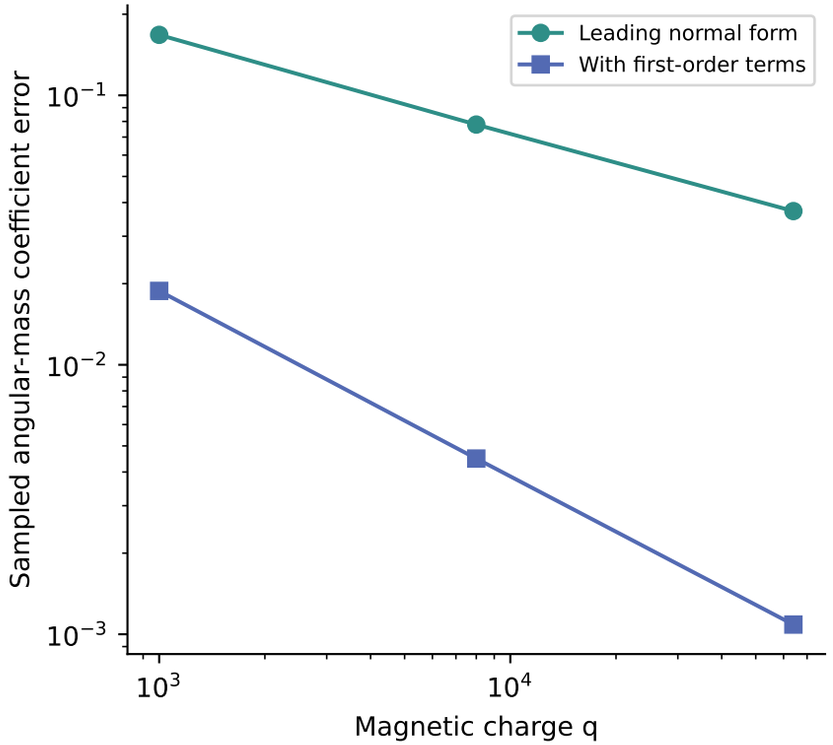
$a - a_* \simeq a_2[(\rho - \rho_*)^2 - z^2]$
Dashed: tangent flux separatrix; squares: test boxes

Label band narrows faster



Bars: $|\eta| \le 1$, width $\propto q^{-2/3}$
Dots: actual nearest half-integer channels

Actual two-dimensional coefficients



Speed and connection errors checked separately.
Local coefficient check; no return probability.